

SCHOOL OF ENGINEERING, CUSAT
Second Series Examination July 2024

Semester : IV

MECHANICAL ENGINEERING (A - Batch)

19-200-0401: COMPLEX VARIABLES AND PARTIAL DIFFERENTIAL EQUATIONS

Time: 2 Hours

Max. Marks: 50

Course Outcomes:

1. Transform a region to another region using conformal mapping
2. Evaluate real integrals using residue theorem
3. Form and solve partial differential equation
4. Determine solution of partial differential equation for vibrating string and heat conduction

BL- Bloom's Taxonomy Levels

(L1- Remember, L2- Understand, L3-Apply, L4-Analyse, L5- Evaluate, L6- Create)

PO- Programme Outcome

Part A
Answer ALL questions

4x5= 20 Marks

Q. No.	Questions	Marks	BL	CO	PO
I a)	Find the singularities of the function $f(z) = \frac{1}{z^4-1}$ and hence find the corresponding residues.	5	L2	2	4
b)	Find the Taylor series expansion of $f(z) = \frac{1}{z}$ about the point i .	5	L5	2	3
c)	Find a complete integral of the equation $pq = 1$.	5	L5	3	2
d)	Derive one-dimensional wave equation for a vibrating string.	5	L1	4	1

Part B

2x15= 30 Marks

		Marks	BL	CO	PO
II	Using contour integration, evaluate $\int_0^{2\pi} \frac{1}{5+4\cos\theta} d\theta$.	10	L5	2	3
OR					
III	State Cauchy's Residue Theorem and use it to evaluate $\int_C \frac{z^2}{z-2}$ where $C: z = 3$.	10	L5	2	3
IV a)	Solve: $\frac{p}{x^2} + \frac{q}{y^2} = z$.	5	L3	3	3
b)	Solve: $(y-z)p + (z-x)q = x-y$.	5	L3	3	3

OR

- | | | | | | | |
|---|----|---|---|----|---|---|
| V | a) | Solve: $r - 4s + 4t = e^{2x+y}$. | 5 | L3 | 3 | 3 |
| | b) | Form the partial differential equation by eliminating the arbitrary function from the relation $f(x^2 + y^2, z - xy) = 0$. | 5 | L2 | 3 | 2 |

- | | | | | | | |
|----|--|--|----|----|---|---|
| VI | | Find the solution of one-dimensional wave equation by the method of separation of variables. | 10 | L1 | 4 | 1 |
|----|--|--|----|----|---|---|

OR

- | | | | | | | |
|-----|--|--|----|----|---|---|
| VII | | Find the solution of one-dimensional heat equation by the method of separation of variables. | 10 | L1 | 4 | 1 |
|-----|--|--|----|----|---|---|

School of Engineering

B.Tech Degree IV semester Examination in Mechanical Engineering,

July 2024

ME1405 Hydraulic Machinery

Time: 2 Hrs

Max. Marks: 65

Test: 2

(A Batch)

Part A

1. a) A hydraulic press has a ram of 300mm diameter and a plunger of 45 mm diameter. Find the weight lifted by the hydraulic press when the force applied at the plunger is 50 N.
- b) Define slip, percentage slip and negative slip of a reciprocating pump.
- c) Describe the working of reciprocating pump with neat sketch.
- d) Show that the force exerted by a jet of water on an inclined fixed plate in the direction of the jet is given by, $F_x = \rho a V^2 \sin^2 \theta$
where a = area of the jet, V = velocity of the jet, and θ = inclination of plate with the jet.

(4 X 3 marks = 12 marks)

PART B

2. A jet of water having a velocity of 15 m/s strikes a curved vane which is moving with velocity of 5 m/s. The vane is symmetrical and is so shaped that the jet is deflected through 120°. Find the angle of the jet at inlet of the vane so that there is no shock. What is the absolute velocity of the jet at outlet in magnitude and direction and the work done per unit weight of water. Assume the vane to be smooth.

(10)

3. A double acting reciprocating pump, running at 50 rpm is discharging 900 litres of water per minute. The pump has stroke of 400 mm. The diameter of piston is 250 mm. The delivery and suction head are 25 m and 4 m respectively. Find the slip of the pump and power required to drive the pump (10)

4. A water turbine has a velocity of 6 m/s at the entrance to the draft tube and a velocity of 1.3 m/s at the exit. For friction losses of 0.1 m and a tail of water 5 m below the entrance to the draft tube, find the pressure head at the entrance. (10)

5. A Kaplan turbine working under a head of 20 m develops 11800kW shaft power. The outer diameter of runner is 3.5 m and hub diameter is 1.75 m. The guide blade angle at the extreme edge of the runner is 35° . The hydraulic and overall efficiencies of the turbines are 88% and 85% respectively. If the velocity of whirl is zero at outlet, determine:

(i) Runner vane angles at inlet and outlet at the extreme edge of the runner, and

(ii) Speed of the turbine

(10)

6. (a). Explain with the help of a neat sketch, the principle and working of the following hydraulic devices: (a) Hydraulic lift (b) Hydraulic crane and (c) Hydraulic torque converter. (7)

(b). Explain with the help of a neat sketch, the principle and working of the following hydraulic devices: (a) Hydraulic intensifier (b) Hydraulic accumulator and (c) Hydraulic ram (6)

DIVISION OF MECHANICAL ENGINEERING, SoE, CUSAT

Time: 2hrs

ME19-200-0403 MECHATRONICS

Marks:50

Internal Test-2 (July 2024)

Course Outcomes (CO):

On completion of the course, the student will be able to

CO1: **Understand** the basics of sensors and transducers and different mechatronics systems.

CO2: **Model** first and second order systems using block diagrams and governing equations.

CO3: **Analyse** first and second order systems for their time and frequency domain responses and stability.

CO4: **Design** mechatronic systems for simple application.

CO5: **Use** software tools for modelling and simulation of mechatronic systems

Programme Outcomes (PO)

PO1-Engineering Knowledge	PO2-Problem Analysis	PO3-Development of solutions	PO4- Conduct investigation of complex problems
PO5- Modern tool usage	PO6- Engineer& Society	PO7-Environment & Sustainability	PO8-Ethics
PO9-Individual& team work	PO10-Communication	PO11-Project management &finance	PO12-Lifelong learning

Programme Educational Objectives (PEO)

PEO1	Preparation: To prepare the graduates or to succeed in industry/technical profession through global rigorous education.
PEO2	Core Competence: To provide the graduates with a solid foundation in mathematical, scientific, and engineering fundamentals required to solve engineering problems and also to pursue higher studies
PEO3	Breadth: To train the graduates with good scientific and engineering breadth so as to comprehend, analyse, design, and create novel solutions and products for the real-life problems
PEO4	Professionalism: To inculcate in the graduates, a professional and ethical attitude, effective communication and team work skills, a multi-disciplinary approach, and an ability to relate engineering issues to broader social context
PEO5	Learning environment: To provide the graduates with an academic environment making them aware of excellence, leadership and teamwork skills, ethical codes and guidelines, and life-long learning needed for a successful professional

Programme Specific Outcomes (PSO)

PSO1	The graduate should excel in product design, thermal and fluid engineering, and manufacturing systems with the acquired knowledge in mathematics, science, and engineering disciplines.
PSO2	The graduate should be competent to analyse, interpret, and solve real life mechanical engineering problems
PSO3	The graduate should have the managerial skills to work effectively in a team and in a society by following ethical and environmental practices

Mapping of Course Outcomes with Programme Outcomes and Programme Specific Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	1	1				1				1	3	3	
CO2	3	3	1	3	1			1				1	3	3	
CO3	3	3	1	3	1			1				1	3	3	
CO4	3	3	1	1	2			1				1	3	3	
CO5	1	1	1	1	3			1				1	3	3	

Bloom's Taxonomy Levels (BL)

Remember	Understand	Apply	Analyse	Design	Create
L1	L2	L3	L4	L5	L6

Mapping of Questions with Course Outcomes

Mapping of Questions with Blooms Taxonomy levels

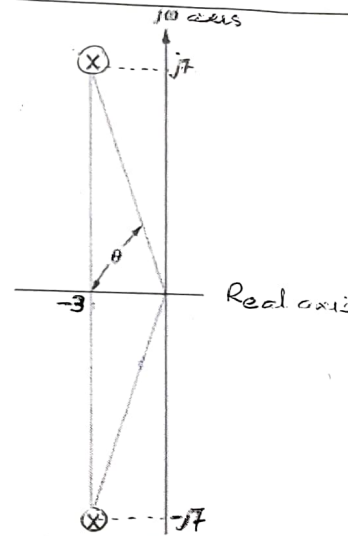
	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Q1	4		3	3			4	3	3		
Q2		10						10			
Q3			10						10		
Q4			10						10		
Q5				10						10	
Total	4	10	23	13			4	13	23	10	
%	8%	20%	46%	26%			8%	26%	46%	20%	

(PTO)

PART A

Question

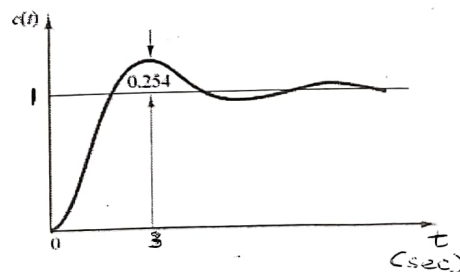
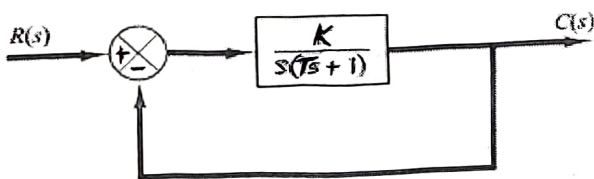
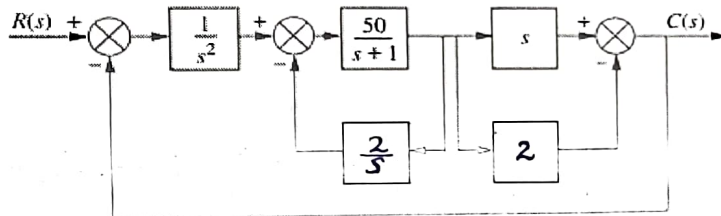
	Question	CO	BL	Marks
a	Differentiate between a microprocessor and a microcontroller	CO1	L2	4
b	The location of poles of a second order system is shown in figure. What is the transfer function of the system	CO3	L4	3
c	Represent the following Boolean algebra equation in a PLC logic ladder diagram $Z = A \cdot \bar{B} \cdot C + \bar{D} + E$	CO2	L3	3



PART B

Question

	Question	CO	BL	Marks
1	Reduce the given block diagram to a single block and find the overall transfer function of the system (either by block diagram reduction or by signal flow graph).	CO2	L3	10
2	A unity negative feedback control system has the plant $G(s) = \frac{50}{s(s+10)}$. Determine the percentage overshoot, 2% settling time x , rise time, and peak time a unit step input	CO3	L4	10
3	When the system shown below is subjected to a unit step input, the response is like the figure shown on right side. Determine the value of K and T	CO3	L4	10
4	A washing machine has a pump to pump in water, a heater to heat water and another pump-to-pump water out. Write the ladder program to achieve the following: The pump is switched on to fill water for 3 minutes. Then the water is heated till the required temperature is reached, i.e., till the temperature sensor produce signal. On pressing the switch for the second pump, it pumps out water for 2 minutes and stops	CO4	L5	10



SCHOOL OF ENGINEERING, CUSAT
Second Series Examination July 2024

Semester : IV

MECHANICAL ENGINEERING

A19-205-0402:METROLOGY & INSTRUMENTATION

Time: 2 Hours

Max. Marks: 50

Course Outcomes:

1. Understand the concepts of measurements, limits, fits and tolerance.
2. Learn about measurement techniques of angles, tapers, and surface finish
3. Learn about applications of measuring instruments
4. Understand the measurement techniques for strain, force, torque, temperature, air pollution and nuclear radiation.

BL- Bloom's Taxonomy Levels

(L1- Remember, L2- Understand, L3-Apply, L4-Analyse, L5- Evaluate, L6- Create)

PO- Programme Outcome

		Part A		4x5= 20 Marks			
		Answer ALL questions					
Q. No.		Questions	Marks	BL	CO	PO	
I	a)	Explain the principle of sine bar with neat sketch.	5	L1	2	1,2	
	b)	Sketch and explain vernier bevel protractor.	5	L1	2	1,2	
	c)	Explain the working of NPL flatness type interferometer with the help of a neat sketch.	5	L1	2	1,2	
	d)	The output of a system (y) is given by the following equation: $y = ax + b$ where x is the input and a and b are constants. Explain the system behavior(linear/non-linear) and explain why is it so?	5	L2	3	1,2	
			2X15 = 30 Marks				
		Part B	Marks	BL	CO	PO	
II	a)	Do the modelling of a liquid in glass thermometer measurement system. What are the assumptions made during modelling. What is the order of the system.	10	L3	3	1,2	
	b)	A thermometer has a time constant of 3.5 s. It is quickly taken from a temperature of 0°C to a water having a temperature of 100°C. What temperature will be indicated in the thermometer after 1.5s. Write down the assumptions, if any, made.	5	L3	3	1,2	

		OR				
III		Explain (i) Strain rosette (ii) Gauge factor (iii) Proving ring	15	L1	4	1,2
IV		Explain any two types of strain gauges with sketch .	15	L1	4	1,2
		OR				
V		Explain with sketches a multi lever platform scale for force measurements.	15	L1	4	1,2

Time: 2hrs	Department of Mechanical Engineering, SOE 2 nd Internal Examination of S-4 ME-A Batch - July 2024 19-205-0406 MANUFACTURING PROCESSES	Max Marks:40
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Course Outcomes:

On completion of this course the student will be able to:

1. Learn about the sand casting process.
2. Understand the special casting process and casting defects.
3. Learn about different metal forming processes.
4. Understand different metal joining processes and their characteristics.

Part-A (Answer all questions)

(Max marks= 10)

Q No

1. Compare welding with riveting and casting.
2. What are electrodes used in arc welding? Discuss its types.
3. With a neat sketch, explain the parts (nomenclature) of a weld.
4. Differentiate between fusion welding and solid state welding with examples.

CO	BL	Marks
CO4	L2	2.5
CO4	L2	2.5
CO4	L2	2.5
CO4	L2	2.5

Part-B (Answer any 3 questions)

(Max marks= 30)

5. Explain in detail oxyacetylene gas welding with neat sketch. Also explain the types of flames with their applications.
6. Discuss in detail the defects in weldments with appropriate sketches.
7. Write short notes on Soldering and Brazing. Also compare between Welding, Soldering and Brazing.
8. Write short notes on:
 - (a) Thermit welding
 - (b) Metal Inert Gas Welding (MIG)
 - (c) Resistance Welding

CO4	L2	10
CO4	L2	10
CO4	L2	10
CO4	L2	10

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II Internal Examination

IV Semester A Batch

ME 19-205-0404 APPLIED THERMODYNAMICS

(2019 Scheme)

Time: 2 Hours

Max. Marks: 40

(Use of Steam Table is permitted)

(Data not given may be suitably assumed. All such assumptions must be clearly stated)

No	Questions	Marks	BL	CO	PO
1	Write a short note on velocity compounding of an impulse turbine	5	L1	4	1
2	A mass of 1.5 kg of air is compressed in a quasi-static process from 0.1 MPa to 0.7 MPa for which $p v = \text{constant}$. The initial density of air is 1.16 kg/m ³ . Find the work done by the piston to compress the air	10	L5	2	1,2,4
3	A Rankine cycle works between pressures 28 bar and 0.05 bar, the initial condition of steam being dry saturated. Calculate the cycle efficiency, work ratio and specific steam consumption.	10	L4	3	1,2,4
4	Dry saturated steam at a pressure of 12 bar flows through the nozzles at the rate of 4.5 kg/s and discharges at a pressure of 1 bar. The loss due to friction occurs only in the diverging portion of the nozzle and its magnitude is 10% of the total isentropic enthalpy drop. Determine the cross sectional area at the throat and exit of the nozzle	10	L4	4	1,2
5	Explain Wilson line in connection with super saturated flow in nozzles	5	L2	4	1,2